

EDA Notebook

0. Setup Environment

```
In [1]: # DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipynrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
12.9/12.9 MB 45.1 MB/s eta 0:00:00
4.9/4.9 MB 62.2 MB/s eta 0:00:00
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

Mounted at /content/gdrive

You can now save your data files in: /content/gdrive/MyDrive/36106/assignment/AT3/data

Student Information

```
In [ ]: group_name = ""
student_name = "Yi An Tsai"
student_id = "25532767"
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)
```

0. Python Packages

0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
In [ ]: # <Student to fill this section>
```

0.b Import Packages

```
In [2]: import pandas as pd
import altair as alt
```

B. Data Understanding

```
In [3]: # Do not modify this code
try:
    df = pd.read_csv(at.folder_path / "customer.csv")
except Exception as e:
    print(e)
```

B.1 Explore Dataset

```
In [ ]: df.head()
```

```
Out [ ]:
```

	customer_id	person_id	store_id	territory_id	account_number
0	4b5762b3-b7bf-4716-9bf9-5c1c9e1f375d	092707ff-4088-4416-ac86-14491f3120c2	NaN	0ad4c625-bb65-4376-8a41-0a65719b0db8	AW00019475
1	26bfba89-84b7-4b5b-9bab-caf7a2791006	fe043525-e0af-4578-a4b9-bfa7be28f4e4	NaN	25d73ad2-9e13-410b-8b0d-5f26e2b9e4f2	AW00022390
2	cf2cf4cc-fcce-4fe5-9400-78373420b525	ee25bbaf-5cdd-4ed4-92fe-7136c4e803a9	NaN	0ad4c625-bb65-4376-8a41-0a65719b0db8	AW00016320
3	83f558b8-bdab-4781-b112-55d2c275ff52	f0452f05-3690-4170-9fb0-fbf9995bbea7	NaN	2ac923e9-3043-4f5e-bae0-ad28089bf187	AW00026149
4	5edc53f4-7821-4a9b-9c27-a45dd2af713e	16c2532d-adb6-4a33-b0ea-71b87d83a904	NaN	0ad4c625-bb65-4376-8a41-0a65719b0db8	AW00021849

```
In [ ]: df.shape
```

```
Out [ ]: (14275, 5)
```

```
In [ ]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14275 entries, 0 to 14274
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customer_id      14275 non-null  object
1   person_id        14038 non-null  object
2   store_id         452 non-null    object
3   territory_id     14275 non-null  object
4   account_number   14275 non-null  object
dtypes: object(5)
memory usage: 557.7+ KB
```

```
In [ ]: df['store_id'].isnull().sum()
```

```
Out [ ]: np.int64(13823)
```

```
In [ ]: df['person_id'].isnull().sum()
```

```
Out [ ]: np.int64(237)
```

```
In [11]: # the unique values
for i in df.columns:
    print(f"Number of unique values in {i} : {len(df[i].unique())}")
```

```
Number of unique values in customer_id : 10331
Number of unique values in person_id : 9233
Number of unique values in store_id : 161
Number of unique values in territory_id : 4
Number of unique values in account_number : 9392
```

```
In [ ]: dataset_insights = """
```

```
There are 14275 rows and 5 columns in the dataset.

14038 missing values are records in the store_id column.
237 missing values are records in the person_id column.
Other three columns are all complete.

"""
```

```
In [ ]: # Do not modify this code
print_tile(size="h3", key='dataset_insights', value=dataset_insights)
```

B.2 Explore Feature of Interest person_id

```
In [ ]: feature_1_insights = """
No features of interest here because the characteristic of all features represent unique values of customer information.
"""
```

```
In [ ]: # Do not modify this code
print_tile(size="h3", key='feature_1_insights', value=feature_1_insights)
```